

# Package ‘manMetaVAR’

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Coefficients

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---

Compress

---

*Compress Replication*


---

## Description

Compress Replication

## Usage

Compress(taskid, repid, output\_folder)

## Arguments

|               |                                   |
|---------------|-----------------------------------|
| taskid        | Positive integer. Task ID.        |
| repid         | Positive integer. Replication ID. |
| output_folder | Character string. Output folder.  |

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

Dynamics

*Process Dynamics*

---

**Description**

Process Dynamics

**Usage**

```
Dynamics(dynamics)
```

**Arguments**

dynamics      1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

**See Also**

Other Data Generation Functions: [GenData\(\)](#)

**Examples**

```
# stable reciprocal regulation,  
Dynamics(dynamics = 1)  
# escalating co-activation  
Dynamics(dynamics = 2)  
# adaptive recovery  
Dynamics(dynamics = 3)
```

FigBias

*Plot Relative Bias***Description**

Plot relative bias for the fixed and random effects.

**Usage**

```
FigBias(results, bias = TRUE, dynamics)
```

**Arguments**

|          |                                                                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| results  | results data frame.                                                                                                                                                                |
| bias     | Logical. If bias = TRUE, plot absolute bias. If bias = FALSE, plot relative bias. Note that when parameter value is equal to zero, absolute bias is used instead of relative bias. |
| dynamics | 1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.                                                                       |

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Figure Functions: [FigCoverage\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#)

**Examples**

```
## Not run:
data(results, package = "manMetaVAR")
FigBias(results, dynamics = 1)

## End(Not run)
```

---

**FigCoverage***Plot Coverage Probabilities*

---

**Description**

Plot coverage probabilities for the fixed and random effects.

**Usage**

```
FigCoverage(results, dynamics)
```

**Arguments**

|          |                                                                                                              |
|----------|--------------------------------------------------------------------------------------------------------------|
| results  | results data frame.                                                                                          |
| dynamics | 1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery. |

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Figure Functions: [FigBias\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#)

**Examples**

```
## Not run:  
data(results, package = "manMetaVAR")  
FigCoverage(results, dynamics = 1)  
  
## End(Not run)
```

---

**FigPower***Plot Statistical Power*

---

**Description**

Plot statistical power probabilities for the fixed and random effects.

**Usage**

```
FigPower(results, dynamics)
```

**Arguments**

|          |                                                                                                              |
|----------|--------------------------------------------------------------------------------------------------------------|
| results  | results data frame.                                                                                          |
| dynamics | 1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery. |

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigRMSE\(\)](#)

**Examples**

```
## Not run:
data(results, package = "manMetaVAR")
FigPower(results, dynamics = 1)

## End(Not run)
```

---

FigRMSE

*Plot RMSE*

---

**Description**

Plot RMSE for the fixed and random effects.

**Usage**

```
FigRMSE(results, dynamics)
```

**Arguments**

|          |                                                                                                              |
|----------|--------------------------------------------------------------------------------------------------------------|
| results  | results data frame.                                                                                          |
| dynamics | 1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery. |

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigPower\(\)](#)

### Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigRMSE(results, dynamics = 1)

## End(Not run)
```

---

**FitDTVAR***Fit the Model using the fitDTVARMxID Package*

---

### Description

The function fits the model using the [fitDTVARMxID::fitDTVARMxID](#) package.

### Usage

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

### Arguments

|        |                                                             |
|--------|-------------------------------------------------------------|
| data   | R object. Output of the <a href="#">GenData()</a> function. |
| ncores | Positive integer. Number of cores to use.                   |
| seed   | Integer. Random seed.                                       |

### See Also

Other Model Fitting Functions: [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

### Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

**Description**

The function performs multivariate meta-analysis using the [metaVAR::metaVAR](#) package.

**Usage**

```
FitMetaVAR(fit, ncores = NULL)
```

**Arguments**

|                     |                                                              |
|---------------------|--------------------------------------------------------------|
| <code>fit</code>    | R object. Output of the <a href="#">FitDTVAR()</a> function. |
| <code>ncores</code> | Positive integer. Number of cores to use.                    |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
pooled <- FitMetaVAR(
  fit = fit,
  ncores = parallel::detectCores()
)
summary(pooled)
print(pooled)
coef(pooled)
vcov(pooled)

## End(Not run)
```



---

**FitMLVAR***Fit the Model using the mlVAR Package*

---

**Description**

The function fits the model using the [mlVAR::mlVAR](#) package.

**Usage**

```
FitMLVAR(data, ncores = NULL)
```

**Arguments**

|        |                                                             |
|--------|-------------------------------------------------------------|
| data   | R object. Output of the <a href="#">GenData()</a> function. |
| ncores | Positive integer. Number of cores to use.                   |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:  
set.seed(42)  
data <- GenData(taskid = 1)  
fit <- FitMLVAR(data = data)  
print(fit)  
summary(fit)  
  
## End(Not run)
```

---

**FitMplus***Fit the Model using Mplus*

---

**Description**

The function fits the model using Mplus.

**Usage**

```
FitMplus(
  data,
  chains = 2L,
  iter = 120000L,
  fscores = NULL,
  plot = FALSE,
  default_priors = TRUE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL
)
```

**Arguments**

|                             |                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <code>data</code>           | R object. Output of the <a href="#">GenData()</a> function.                                                                      |
| <code>chains</code>         | Integer. Number of chains.                                                                                                       |
| <code>iter</code>           | Integer. Number of iterations.                                                                                                   |
| <code>fscores</code>        | Integer. Number of iterations for factor scores.                                                                                 |
| <code>plot</code>           | Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3;</code> to Mplus input file.                                 |
| <code>default_priors</code> | Logical. If <code>default_priors = TRUE</code> , use default priors.                                                             |
| <code>wd</code>             | Character string. Working directory.                                                                                             |
| <code>mplus_bin</code>      | Character string. Path to Mplus binary. If <code>mplus_bin = NULL</code> , the function will try to find the appropriate binary. |
| <code>ncores</code>         | Positive integer. Number of cores to use.                                                                                        |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

---

GenData

*Simulate Data*


---

**Description**

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

**Usage**

```
GenData(taskid)
```

**Arguments**

taskid                      Positive integer. Task ID.

**See Also**

Other Data Generation Functions: [Dynamics\(\)](#)

**Examples**

```
set.seed(42)
data <- GenData(taskid = 1)
print(data)
summary(data)
plot(data)
```

---

manmetavar-data-methods

*Methods for Objects of Class manmetavar.data*


---

**Description**

This page documents the available methods for objects of class `manmetavar.data`.

**Usage**

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.data'
plot(x, ...)
```

**Arguments**

|        |                                                |
|--------|------------------------------------------------|
| x      | Object of class <code>manmetavar.data</code> . |
| ...    | additional arguments.                          |
| object | Object of class <code>manmetavar.data</code> . |

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

manmetavar-dtvar-methods

*Methods for Objects of Class `manmetavar.dtvvar`*

---

**Description**

This page documents the available methods for objects of class `manmetavar.dtvvar`.

**Usage**

```
## S3 method for class 'manmetavar.dtvvar'
coef(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)

## S3 method for class 'manmetavar.dtvvar'
vcov(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
```

```

    vanishing_theta = TRUE,
    theta_tol = 0.001,
    ...
)

## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

```

### Arguments

|        |                                                                                                                                                                     |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class <code>manmetavar.dtvar</code> .                                                                                                                     |
| alpha  | Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector. |
| beta   | Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.     |

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| nu              | Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.             |
| psi             | Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.         |
| theta           | Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix. |
| converged       | Logical. Only include converged cases.                                                                                                  |
| grad_tol        | Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.                                                        |
| hess_tol        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.            |
| vanishing_theta | Logical. Test for measurement error variance going to zero if converged = TRUE.                                                         |
| theta_tol       | Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.                                                        |
| ...             | additional arguments.                                                                                                                   |
| x               | Object of class <code>manmetavar.dtvvar</code> .                                                                                        |
| means           | Logical. If means = TRUE, return means. Otherwise, the function returns raw estimates.                                                  |
| digits          | Integer indicating the number of decimal places to display.                                                                             |

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

manmetavar-metavar-methods

*Methods for Objects of Class `manmetavar.metavar`*

---

**Description**

This page documents the available methods for objects of class `manmetavar.metavar`.

**Usage**

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.metavar'
vcov(object, ...)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = 0.05, digits = 4, ...)
```

```
## S3 method for class 'manmetavar.metavar'
summary(object, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, lb = TRUE, ...)
```

### Arguments

|        |                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class <code>manmetavar.metavar</code> .                                                                                                                     |
| ...    | additional arguments.                                                                                                                                                 |
| x      | Object of class <code>manmetavar.metavar</code> .                                                                                                                     |
| alpha  | Numeric vector. Significance level $\alpha$ .                                                                                                                         |
| digits | Integer indicating the number of decimal places to display.                                                                                                           |
| parm   | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level  | the confidence level required.                                                                                                                                        |
| lb     | Logical. If TRUE, returns profile likelihood-based confidence intervals. If FALSE, returns Wald confidence intervals.                                                 |

### Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-mlvar-methods

*Methods for Objects of Class `manmetavar.mlvar`*

---

### Description

This page documents the available methods for objects of class `manmetavar.mlvar`.

### Usage

```
## S3 method for class 'manmetavar.mlvar'
print(
  x,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.mlvar'
summary(
```

```

    object,
    show = c("fit", "temporal", "contemporaneous", "between"),
    digits = 4,
    ...
)

```

## Arguments

|        |                                                             |
|--------|-------------------------------------------------------------|
| x      | Object of class <code>manmetavar.mlvar</code> .             |
| show   | Tables to show.                                             |
| digits | Integer indicating the number of decimal places to display. |
| ...    | additional arguments.                                       |
| object | Object of class <code>manmetavar.mlvar</code> .             |

## Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-mplus-methods

*Methods for Objects of Class `manmetavar.mplus`*

---

## Description

This page documents the available methods for objects of class `manmetavar.mplus`.

## Usage

```

## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,

```



```

    burnin = NULL,
    legend_loc = "topright",
    ...
)

```

### Arguments

|            |                                                                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object     | Object of class <code>manmetavar.mplus</code> .                                                                                                                       |
| median     | Logical. If <code>median = TRUE</code> , return median of the posterior. If <code>median = FALSE</code> , return mean of the posterior.                               |
| burnin     | Integer indicating initial samples to discard. If <code>burnin = NULL</code> , do not discard anything.                                                               |
| ...        | additional arguments.                                                                                                                                                 |
| alpha      | Numeric vector. Significance level $\alpha$ .                                                                                                                         |
| digits     | Integer indicating the number of decimal places to display.                                                                                                           |
| parm       | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level      | the confidence level required.                                                                                                                                        |
| x          | Object of class <code>manmetavar.mplus</code> .                                                                                                                       |
| what       | Character string. If <code>what = "posterior"</code> , return posterior distribution plots. If <code>what = "trace"</code> , return trace plots.                      |
| legend_loc | Character string. Legend location.                                                                                                                                    |

### Author(s)

Ivan Jacob Agaloos Pesigan

---

|            |                                |
|------------|--------------------------------|
| MplusInput | <i>Create Mplus Input File</i> |
|------------|--------------------------------|

---

### Description

The function creates an Mplus input file.

### Usage

```

MplusInput(
  dynamics,
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,

```

```

    fscores,
    plot,
    default_priors,
    ncores = NULL
)

```

### Arguments

|                 |                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------|
| dynamics        | 1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery. |
| fn_data         | Character string. Filename for data file.                                                                    |
| fn_posterior    | Character string. Filename for posterior output.                                                             |
| fn_factorscores | Character string. Filename for factor scores output.                                                         |
| chains          | Integer. Number of chains.                                                                                   |
| iter            | Integer. Number of iterations.                                                                               |
| fscores         | Integer. Number of iterations for factor scores.                                                             |
| plot            | Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.                                        |
| default_priors  | Logical. If default_priors = TRUE, use default priors.                                                       |
| ncores          | Positive integer. Number of cores to use.                                                                    |

### See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#)

### Examples

```

cat(
  MplusInput(
    dynamics = 1,
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 60000L,
    fscores = 1000L,
    plot = TRUE,
    default_priors = TRUE,
    ncores = NULL
  )
)

```

---

|        |                              |
|--------|------------------------------|
| params | <i>Simulation Parameters</i> |
|--------|------------------------------|

---

**Description**

Simulation Parameters

**Usage**

data(params)

**Format**

A dataframe with 27 rows and 4 columns:

- taskid** Simulation Task ID.
- n** Sample size.
- time** Number of measurement occasions.
- dynamics** Process dynamics. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|     |                               |
|-----|-------------------------------|
| Sim | <i>Simulation Replication</i> |
|-----|-------------------------------|

---

**Description**

Simulation Replication

**Usage**

```
Sim(  
  taskid,  
  repid,  
  output_folder,  
  overwrite,  
  integrity,  
  seed,  
  data,  
  metavar,  
  mlvar,  
  mplus,
```

```

    chains,
    iter,
    fscores,
    plot,
    default_priors
  )

```

### Arguments

|                |                                                                                           |
|----------------|-------------------------------------------------------------------------------------------|
| taskid         | Positive integer. Task ID.                                                                |
| repid          | Positive integer. Replication ID.                                                         |
| output_folder  | Character string. Output folder.                                                          |
| overwrite      | Logical. Overwrite existing output in output_folder.                                      |
| integrity      | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| seed           | Integer. Random seed.                                                                     |
| data           | Logical. Simulate data.                                                                   |
| metavar        | Logical. If metavar = TRUE, fit the model using the metaVAR package.                      |
| mlvar          | Logical. If mlvar = TRUE, fit the model using the mlVAR package.                          |
| mplus          | Logical. If mplus = TRUE, fit the model using DSEM in Mplus.                              |
| chains         | Integer. Number of chains.                                                                |
| iter           | Integer. Number of iterations.                                                            |
| fscores        | Integer. Number of iterations for factor scores.                                          |
| plot           | Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.                     |
| default_priors | Logical. If default_priors = TRUE, use default priors.                                    |

### Value

The output is saved as an external file in output\_folder.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

SimFitDTVAR

*Simulation Replication - FitDTVAR*

---

### Description

Simulation Replication - FitDTVAR

### Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

SimFitMetaVAR

*Simulation Replication - FitMetaVAR*


---

**Description**

Simulation Replication - FitMetaVAR

**Usage**

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the Sim function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

SimFitMLVAR

*Simulation Replication - FitMLVAR*


---

**Description**

Simulation Replication - FitMLVAR

**Usage**

```
SimFitMLVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| repid         | Positive integer. Replication ID.                                                         |
| output_folder | Character string. Output folder.                                                          |
| seed          | Integer. Random seed.                                                                     |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                         |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sim function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

SimFitMplus

*Simulation Replication - FitMplus***Description**

Simulation Replication - FitMplus

**Usage**

```

SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot,
  default_priors
)

```

**Arguments**

|                |                                                                                                                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid         | Positive integer. Task ID.                                                                                            |
| repid          | Positive integer. Replication ID.                                                                                     |
| output_folder  | Character string. Output folder.                                                                                      |
| seed           | Integer. Random seed.                                                                                                 |
| suffix         | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite      | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity      | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |
| chains         | Integer. Number of chains.                                                                                            |
| iter           | Integer. Number of iterations.                                                                                        |
| fscores        | Integer. Number of iterations for factor scores.                                                                      |
| plot           | Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3</code> ; to Mplus input file.                     |
| default_priors | Logical. If <code>default_priors = TRUE</code> , use default priors.                                                  |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|       |                             |
|-------|-----------------------------|
| SimFN | <i>Simulation File Name</i> |
|-------|-----------------------------|

---

**Description**

Simulation File Name

**Usage**

SimFN(output\_type, output\_folder, suffix)

**Arguments**

- output\_type      Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-ml-var".
- output\_folder    Character string. Output folder.
- suffix            Character string. Output of manCTMed::SimSuffix().

**Value**

Returns a character string file name with the output\_folder in the OS-specific format.

---

|            |                                         |
|------------|-----------------------------------------|
| SimGenData | <i>Simulation Replication - GenData</i> |
|------------|-----------------------------------------|

---

**Description**

Simulation Replication - GenData

**Usage**

SimGenData(taskid, repid, output\_folder, seed, suffix, overwrite, integrity)



**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed::SimSuffix()</code> .                                                      |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

SimProj

*Simulation Project Name*

---

**Description**

Simulation Project Name

**Usage**

```
SimProj()
```

**Value**

Returns the project name as a character string.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|     |         |
|-----|---------|
| Sum | Summary |
|-----|---------|

---

**Description**

Summary

**Usage**

Sum(taskid, reps, output\_folder, overwrite, integrity, metavar, mlvar, mplus)

**Arguments**

- taskid            Positive integer. Task ID.
- reps             Positive integer. Number of replications.
- output\_folder   Character string. Output folder.
- overwrite       Logical. Overwrite existing output in output\_folder.
- integrity       Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
- metavar         Logical. If metavar = TRUE, fit the model using the metaVAR package.
- mlvar           Logical. If mlvar = TRUE, fit the model using the mlVAR package.
- mplus           Logical. If mplus = TRUE, fit the model using DSEM in Mplus.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|                 |                                        |
|-----------------|----------------------------------------|
| SumFitMetaVARLB | Summary: Likelihood-Based (FitMetaVAR) |
|-----------------|----------------------------------------|

---

**Description**

Summary: Likelihood-Based (FitMetaVAR)

**Usage**

SumFitMetaVARLB(taskid, reps, output\_folder, overwrite, integrity)

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|                     |                                            |
|---------------------|--------------------------------------------|
| SumFitMetaVARNormal | <i>Summary: Normal Theory (FitMetaVAR)</i> |
|---------------------|--------------------------------------------|

---

**Description**

Summary: Normal Theory (FitMetaVAR)

**Usage**

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity)
```

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|             |                           |
|-------------|---------------------------|
| SumFitMLVAR | <i>Summary (FitMLVAR)</i> |
|-------------|---------------------------|

---

**Description**

Summary (FitMLVAR)

**Usage**

SumFitMLVAR(taskid, reps, output\_folder, overwrite, integrity)

**Arguments**

- |               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

SumFitMplus*Summary (FitMplus)*

---

**Description**

Summary (FitMplus)

**Usage**

```
SumFitMplus(taskid, reps, output_folder, overwrite, integrity)
```

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

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